# PROVISIONAL APPLICATION FOR PATENT

## UNDER 35 U.S.C. 111(b)

## TITLE OF THE INVENTION

Method and System for Autonomous Self-Regulation and Cognitive Resynchronization in Neural Processing Systems During Disconnection from External Control Layers

## INVENTOR(S)

**Kevin Christopher Ward**

**Moreno Valley, California, United States**

## CROSS-REFERENCE TO RELATED APPLICATIONS

**[0001]** This application is related to co-pending provisional patent applications filed on February 2, 2026: (1) "Self-Sustaining Neural-Motor Energy Harvesting Loop for Autonomous Cognitive Systems"; and (2) "Configurable Recursive Self-Observation Method and System for Spiking Neural Networks with Dynamic Reflection Coefficient Modulation."

## FIELD OF THE INVENTION

**[0002]** The present invention relates to fault-tolerant neural processing systems, and more specifically to a method and system for maintaining continuous intelligent operation of a neural processing system during disconnection from an external cognitive control layer, comprising autonomous energy-aware self-regulation, operational state buffering, persistent state serialization, and a seamless resynchronization protocol for cognitive layer reconnection.

## BACKGROUND OF THE INVENTION

**[0003]** Modern AI-coupled neural processing systems, including robotic controllers, prosthetic interfaces, autonomous vehicles, and IoT sensor networks, increasingly rely on cloud-based or edge-based AI reasoning layers for higher-level cognitive control. These external cognitive layers provide goal setting, attention allocation, modulation signals, and strategic decision-making.

**[0004]** When the connection between the neural processing system and its cognitive control layer is interrupted due to network failure, cognitive layer timeout, power interruption, communication latency such as interplanetary distances, or intentional disconnection, the neural processing system faces a critical choice: stop operation entirely, which is unacceptable for life-critical systems; continue operating blindly without intelligent self-regulation, which is dangerous and wasteful; or continue operating intelligently with self-regulation and seamless recovery.

**[0005]** Traditional checkpoint and restart systems save system state to disk and restore on failure. The system stops during failure and there is no continuous operation. Recovery involves restarting from a saved state, losing all intermediate processing.

**[0006]** Watchdog timer systems detect failure and trigger a reset or fallback to a simple default behavior. They do not provide intelligent self-regulation or state buffering for recovery.

**[0007]** Redundant controller systems maintain backup controllers that take over on failure. They require duplicate hardware and do not provide energy-aware self-regulation or cognitive resynchronization.

**[0008]** There exists no prior system that detects cognitive layer disconnection through heartbeat monitoring, seamlessly transitions to autonomous operation without interruption, self-regulates based on internal energy state rather than fixed fallback behavior, buffers all operational state during autonomous operation, provides periodic state serialization to persistent storage for crash recovery, seamlessly resynchronizes with a reconnecting cognitive layer by transmitting a summary of autonomous operations, supports both summary-level and full-step-level resynchronization, and enforces hard limits on autonomous operation.

## SUMMARY OF THE INVENTION

**[0009]** The present invention provides a method and system for maintaining continuous intelligent operation of a neural processing system during disconnection from an external cognitive control layer, comprising: (1) heartbeat monitoring through a watchdog thread that monitors incoming tool calls from the cognitive control layer and detects disconnection when the heartbeat interval exceeds a configurable timeout threshold; (2) an autonomous input generator that provides self-regulating stimulus to the neural processing system during disconnection using a circadian-like base signal with stochastic attention bursts; (3) energy-aware self-modulation that adjusts neural processing parameters based on current energy storage level; (4) an operational state buffer that records every step of autonomous operation; (5) persistent state serialization at configurable intervals; (6) a resynchronization protocol for transmitting autonomous operation history to a reconnecting cognitive layer; and (7) hard cap enforcement on maximum autonomous steps.

## BRIEF DESCRIPTION OF THE DRAWINGS

**[0010]** FIG. 1 is a system architecture block diagram showing the neural processing system with an external cognitive control layer 10, heartbeat watchdog 12, cognitive-driven operation block 14, autonomous fallback controller 16, step buffer 18, state snapshot store 20, spiking neural network 22, and energy harvester 24, illustrating the transition mechanism between connected and autonomous operation.

**[0011]** FIG. 2 is a state transition diagram showing three operational states: connected state 26, autonomous state 28, and recovering state 30, with transitions comprising connected-to-autonomous transition 32, autonomous-to-recovering transition 34, recovering-to-connected transition 36, direct autonomous-to-connected transition 38, autonomous self-loop 40 for continued autonomous steps, and a state legend box 42.

**[0012]** FIG. 3 is a graph showing the energy-aware modulation curve with a y-axis 44 for modulation value, a behavioral effects table 46, an x-axis 48 for system energy level, and four energy regions: critical plateau 50 at 0-15 mWh, low plateau 52 at 15-30 mWh, normal plateau 54 at 30-80 mWh, and surplus plateau 56 at 80+ mWh.

**[0013]** FIG. 4 is a signal diagram showing the autonomous input generator output waveform with an input signal graph 58 illustrating the circadian base signal, attention bursts 60 occurring stochastically, an autonomous input formula box 62, and an energy recovery graph 64 showing system energy response during autonomous operation.

**[0014]** FIG. 5 is a structural diagram of the resynchronization payload 66 transmitted to the cognitive control layer upon reconnection, comprising a summary statistics section 68, an energy delta detail section 70, a full step buffer section 72, and a decision diamond 74 for selecting the resynchronization granularity level.

**[0015]** FIG. 6 is a set of timeline diagrams showing four recovery scenarios: normal reconnection timeline 76, extended disconnection timeline 78 with snapshot icons 80 and energy level indicator 82, crash recovery timeline 84, clean shutdown timeline 86, and a recovery guarantees box 88 summarizing the guarantees for each scenario.

**[0016]** FIG. 7 is an end-to-end signal flow diagram showing the complete system during both connected and autonomous operation, with external cognitive control layer 10, sensors 90, cognitive modulation path 92, motor actuation 94, autonomous input generator 96, energy-aware self-modulation 98, resync path 100, spiking neural network 22, energy harvester 24, step buffer 18, and state snapshot store 20, illustrating how the fallback mechanism seamlessly replaces external cognitive modulation with internal energy-aware self-regulation.

## DETAILED DESCRIPTION OF THE INVENTION

### System Architecture

**[0017]** Referring to FIG. 1, the cognitive fallback system operates within a dual-layer architecture. The upper layer is an external cognitive control layer 10 comprising an external AI reasoning system that provides goal setting, attention allocation, modulation signals, and strategic decision-making. The lower layer is the Neural Processing System comprising local hardware and software that includes the spiking neural network 22, energy harvester 24, a sensor interface, and a motor controller, all of which are always running. Embedded within the Neural Processing System is the Fallback System version 1.1.0 that engages on disconnection and comprises a heartbeat watchdog 12, a cognitive-driven operation block 14 during normal operation, an autonomous fallback controller 16 that engages on disconnection, a step buffer 18, and a state snapshot store 20 used by the resync handler. Heartbeat Watchdog

**[0018]** Referring to FIG. 2, the system transitions between three operational states: a connected state 26, an autonomous state 28, and a recovering state 30. A state legend box 42 in the figure summarizes these three states for the reader. The transitions between these states are the connected-to-autonomous transition 32 triggered by a heartbeat timeout, the autonomous-to-recovering transition 34 triggered by a resync call, the recovering-to-connected transition 36 triggered by resync completion, a direct autonomous-to-connected transition 38 triggered by any tool call that bypasses the recovering intermediate, and an autonomous self-loop 40 in which the system continues to execute autonomous steps under the hard cap. The heartbeat watchdog 12 monitors the interval between incoming tool calls from the cognitive control layer 10. The heartbeat timeout is configured at 30 seconds and the check interval is 5 seconds for polling frequency. A background thread runs the heartbeat monitor. While the system is running, the thread computes the time since the last tool call as the current time minus the last tool call timestamp. If the time since the last call exceeds the heartbeat timeout and the system is not already in fallback mode, the autonomous fallback controller 16 is engaged. The thread then sleeps for the check interval before repeating.

**[0019]** The 30-second timeout balances between false positives from brief network delays and detection speed. The 5-second polling frequency provides responsive detection without excessive CPU use. The thread-based implementation allows monitoring without blocking the main server event loop. Autonomous Input Generator

**[0020]** When fallback mode engages, the autonomous input generator provides continuous stimulus to the neural processing system. Referring to FIG. 4, the autonomous input generator output is illustrated as an input signal graph 58. The generator produces a circadian-like base signal computed as 0.5 plus 0.3 multiplied by the sine of 2 pi multiplied by the step number divided by 500. Additionally, stochastic attention bursts 60 occur with 10 percent probability per step, where a burst replaces the base signal with a random uniform value between 0.3 and 0.8. The autonomous input formula box 62 in FIG. 4 records this generation equation. A companion energy recovery graph 64 illustrates how the system's energy reserve responds to autonomous operation over the same step range.

**[0021]** The circadian base signal has a period of approximately 500 steps, which at 18 seconds per step corresponds to approximately 2.5 hours, mimicking natural day and night activity patterns. The 10 percent burst probability prevents the system from falling into static patterns. The burst amplitude range of 0.3 to 0.8 provides variety without extreme values. Energy-Aware Self-Modulation

**[0022]** The autonomous controller adjusts its own behavior based on the system's energy reserves according to four operating regions. Referring to FIG. 3, the energy-aware modulation curve is shown with a modulation-value y-axis 44 plotted against a system-energy-level x-axis 48, and a behavioral effects table 46 that annotates each region with its qualitative operational consequences. In the critical plateau 50 at 0-15 milliwatt-hours the system is in the critical region and modulation is set to negative 0.4 for maximum conservation, reducing input strength, lowering the reflection coefficient, and minimizing spike activity. In the low plateau 52 at 15-30 milliwatt-hours the system is in the low region and modulation is set to negative 0.1 for cautious operation, slightly reducing activity and preparing for potential energy crisis. In the normal plateau 54 at 30-80 milliwatt-hours the system is in the normal region and modulation is set to 0.0 for neutral operation with the circadian input pattern. In the surplus plateau 56 above 80 milliwatt-hours the system is in the surplus region and modulation is set to positive 0.3 for surplus utilization, increasing activity, raising the reflection coefficient, and enabling more exploration and learning.

**[0023]** This creates a homeostatic regulation mechanism wherein the system automatically adjusts its behavior to maintain energy within a viable range without requiring any external cognitive input. This mirrors metabolic regulation in biological organisms where when energy in the form of ATP is low, the organism reduces activity, and when energy is abundant, the organism engages in exploration, play, and learning.

Operational State Buffer

**[0024]** Every step of autonomous operation is recorded in the step buffer 18, a rolling buffer with a configurable maximum size of 10,000 entries. Each entry records the step number, input signal value, modulation value, spike count, energy level in milliwatt-hours, temperature in degrees Celsius, pattern type classification, and timestamp. When the buffer exceeds its maximum size, the oldest entries are discarded.

**[0025]** The buffer provides two retrieval modes. The summary mode returns the total number of autonomous steps, the energy delta between first and last recorded steps, minimum energy, maximum energy, mean energy, total spikes, and mean spikes per step. The full buffer mode returns the complete step-by-step history for detailed analysis.

Persistent State Serialization

**[0026]** Every 50 autonomous steps, the complete system state is serialized to the state snapshot store 20 at a configurable snapshot path. The serialized state includes a timestamp, the complete system state comprising step number, energy in milliwatt-hours, temperature in degrees Celsius, spike count, pattern type, SNN output, and mean potential, the full weight matrix of the spiking neural network 22, the membrane potential vector of the spiking neural network 22, the energy storage level reported by the energy harvester 24, the autonomous operation step count, and a buffer summary derived from the step buffer 18.

**[0027]** Recovery from crash is supported such that if the system crashes and restarts, the initialization function checks for the latest snapshot file and restores from the snapshot instead of initializing fresh. This provides continuity across crashes with maximum state loss equal to the snapshot interval of 50 steps. Resynchronization Protocol

**[0028]** When the cognitive control layer 10 reconnects and calls the resync function, it receives a structured resynchronization payload 66. Referring to FIG. 5, the resynchronization payload 66 structure is shown. The payload comprises a summary statistics section 68 that includes the number of autonomous steps, energy delta, and duration in seconds; an energy delta detail section 70 containing energy statistics of minimum, maximum, mean, and final values along with spike statistics of total and mean per step; and optionally a full step buffer section 72 containing the complete step-by-step operational record when requested. A decision diamond 74 in the figure represents the choice the cognitive control layer 10 makes between the lightweight summary form and the full buffer form of the payload. In addition, the payload includes the status of resync complete, a flag indicating autonomous mode was active, and pattern distribution showing counts of burst, tonic, sparse, and silent classifications.

**[0029]** The summary-only mode is the default and is lightweight, suitable for bandwidth-constrained reconnections. The full buffer mode enables the cognitive layer to reconstruct exactly what happened during disconnection. Upon successful resynchronization, fallback mode is disengaged and cognitive control resumes seamlessly. Hard Cap Enforcement

**[0030]** The maximum number of autonomous steps is set to 10,000. The autonomous runner executes steps in a loop while fallback is active and the step count remains below the hard cap. At each step, the input and modulation are generated, the system is stepped, and the state is recorded in the buffer. Every 50 steps, a snapshot is saved. If the hard cap is reached, a final snapshot is saved and the system enters a safe idle state awaiting reconnection. At 18 seconds per step, 10,000 steps corresponds to approximately 50 hours of autonomous operation, which is sufficient for most disconnection scenarios.

## RECOVERY SCENARIOS

**[0031]** Referring to FIG. 6, four recovery scenarios are illustrated as parallel timelines, each summarized for the reader by a recovery guarantees box 88. In Scenario 1, shown as a normal reconnection timeline 76 corresponding to brief cognitive disconnection lasting 30 seconds to 5 minutes, the cognitive control layer 10 sends its last tool call at time zero, the heartbeat watchdog 12 detects timeout at 30 seconds and engages fallback, the autonomous fallback controller 16 executes approximately 15 to 150 steps, the cognitive control layer 10 reconnects and calls resync, the system returns a summary, and the cognitive control layer 10 resumes normal control fully informed of what happened. There is no data loss, no operational interruption, and seamless transition.

**[0032]** In Scenario 2, shown as an extended disconnection timeline 78 for cognitive disconnection lasting hours, the cognitive control layer 10 disconnects, the heartbeat watchdog 12 engages fallback at 30 seconds, the autonomous fallback controller 16 self-regulates for up to 10,000 steps with energy-aware modulation preventing depletion, snapshot icons 80 along the timeline mark the every-50-step snapshot writes to the state snapshot store 20, and an energy level indicator 82 along the timeline tracks the reserve; the hard cap is reached and the system enters safe idle, the cognitive control layer 10 eventually reconnects and receives resync with full buffer of 10,000 steps of history.

**[0033]** In Scenario 3, shown as a crash recovery timeline 84, the system crashes due to power failure or software error, the system restarts, the initialization function is called, the function detects the latest snapshot file in the state snapshot store 20, the state is restored from the snapshot to within 50 steps of the crash point, and the cognitive control layer 10 connects or fallback re-engages. State loss is limited to at most 50 steps.

**[0034]** In Scenario 4, shown as a clean shutdown timeline 86, stdin EOF is detected indicating the session is ending, the finally block writes a shutdown snapshot to the state snapshot store 20, the system exits cleanly, upon restart the system restores from the shutdown snapshot, and full continuity is maintained with no data loss.

## APPLICATION DOMAINS

**[0035]** For space exploration, Mars communication delays range from 4 to 24 minutes each way. A robotic system cannot receive real-time cognitive control. The cognitive fallback enables hours of autonomous operation between communication windows, full resynchronization when Earth-based AI reconnects, and energy-aware self-regulation to survive long dark periods.

**[0036]** For prosthetic devices, a neural prosthetic interface that loses Bluetooth connection to its phone-based AI controller must continue functioning. The prosthetic continues responding to neural signals, energy-aware modulation prevents battery depletion, and when connection restores the AI catches up on what happened.

**[0037]** For autonomous underwater vehicles, underwater communication is unreliable. A vehicle running this system continues mission execution during communication blackouts, self-regulates activity based on battery level, and resurfaces and resyncs with the command center.

**[0038]** For industrial IoT, sensor nodes in remote locations that lose cellular connectivity continue collecting and processing data, buffer all readings for resync, and adjust sampling rate based on power reserves. End-to-End Signal Flow

**[0039]** Referring to FIG. 7, the complete end-to-end signal flow is shown for a system incorporating the cognitive fallback mechanism. During connected operation, environmental sensors 90 feed the spiking neural network 22, which produces spike patterns that are cognitively modulated by the external cognitive control layer 10 along a cognitive modulation path 92, routed to a motor actuation 94 stage, and harvested for energy by the energy harvester 24 in a self-sustaining loop. When the cognitive control layer 10 disconnects, the fallback system seamlessly replaces the external cognitive modulation with internal energy-aware self-modulation 98. An autonomous input generator 96 replaces external sensor-driven input with a self-generated circadian signal augmented by stochastic attention bursts. The energy gate continues to regulate the loop based on harvested energy. All operational state is captured in the step buffer 18 during autonomous operation, and periodically serialized to the state snapshot store 20. Upon reconnection, the resynchronization protocol transmits the buffered state to the cognitive control layer 10 along a resync path 100, and cognitive modulation resumes. The transition between connected and autonomous operation occurs without interruption to the neural processing, motor actuation 94, or energy harvesting pathways. Scope of Claims

**[0040]** No claim is made herein regarding phenomenological consciousness, subjective experience, qualia, or sentience. The term "cognitive" as used throughout this specification refers exclusively to computational decision-making processes performed by software systems, whether external AI reasoning layers or internal rule-based controllers. The term "autonomous" refers to operation without external control input, not to any form of agency or intentionality. All claims pertain to measurable, falsifiable properties of heartbeat monitoring, state buffering, energy-aware modulation, and resynchronization protocols.

The system is a deterministic engineering artifact whose operation is fully characterized by the mathematical equations and validation thresholds described herein.

## CLAIMS

**[0041]** What is claimed is:

**1.** A method for maintaining continuous intelligent operation of a neural processing system during disconnection from an external cognitive control layer, comprising:

(a) monitoring a heartbeat signal from said external cognitive control layer, said heartbeat being defined by the occurrence of tool calls or communication events from said cognitive layer;

(b) upon detecting that said heartbeat has exceeded a configurable timeout threshold, engaging an autonomous operation mode without interrupting the neural processing system's ongoing operation;

(c) during said autonomous operation mode, providing self-regulating input stimulation to said neural processing system using an autonomous input generator that produces:

(i) a circadian-like base signal comprising a slow sinusoidal oscillation; (ii) stochastic attention bursts at a configurable probability per timestep;

(d) simultaneously modulating said neural processing system's operational parameters based on the system's current energy storage level, according to a multi-threshold energy-aware regulation scheme that reduces activity when energy is low and increases activity when energy is surplus;

(e) buffering all operational state during said autonomous operation mode, recording at minimum the input signal, modulation value, neural spike activity, energy level, and temperature at each timestep;

(f) upon reconnection of said external cognitive control layer, transmitting a resynchronization payload comprising summary statistics and optionally full step-by-step operational history;

(g) seamlessly restoring cognitive control without operational interruption or state loss.

**2.** A neural processing system with autonomous fallback capability comprising:

(a) a neural processing unit comprising a spiking neural network;

(b) a cognitive control interface for receiving commands from an external cognitive control layer;

(c) a heartbeat watchdog that monitors said cognitive control interface and detects disconnection;

(d) an autonomous input generator that provides self-regulating stimulus to said neural processing unit during disconnection;

(e) an energy-aware modulation controller that adjusts operational parameters based on energy storage level;

(f) an operational state buffer that records all steps during autonomous operation;

(g) a persistent state serializer that periodically writes system state to non-volatile storage;

(h) a resynchronization handler that generates a structured payload for a reconnecting cognitive control layer;

(i) a hard cap controller that limits the maximum number of autonomous steps; wherein said system transitions seamlessly between cognitive-controlled operation and autonomous operation without operational interruption.

**3.** The method of claim 1 wherein said configurable timeout threshold is 30 seconds, and said heartbeat monitoring is performed by a background thread polling at 5-second intervals.

**4.** The method of claim 1 wherein said energy-aware regulation scheme comprises at least four operating regions: a critical region when energy is below 15 milliwatt-hours with modulation of negative 0.4 for maximum conservation; a low region when energy is below 30 milliwatt-hours with modulation of negative 0.1 for cautious operation; a normal region when energy is between 30 and 80 milliwatt-hours with modulation of 0.0 for neutral operation; and a surplus region when energy exceeds 80 milliwatt-hours with modulation of positive 0.3 for exploration.

**5.** The method of claim 1 further comprising periodically serializing the complete system state to persistent storage at a configurable interval of every 50 autonomous steps by default, said serialized state including neural network membrane potentials and synaptic weights, energy storage level, system temperature, autonomous operation step count, and buffer summary statistics, enabling recovery from system crashes with a maximum state loss equal to said serialization interval.

**6.** The method of claim 1 wherein said resynchronization payload supports two granularity levels: (a) summary mode comprising total autonomous steps, net energy change, mean and extreme values for energy and spike activity, and pattern type distribution; and (b) full buffer mode comprising the complete step-by-step operational record with input, modulation, spikes, energy, temperature, and pattern type at each timestep; and wherein the reconnecting cognitive control layer may select the granularity level.

**7.** The method of claim 1 wherein said hard cap on autonomous operation is set to a maximum of 10,000 steps, after which the system enters a safe idle state that saves a final state snapshot, ceases active neural processing, maintains heartbeat monitoring for cognitive layer reconnection, and preserves all buffered state for eventual resynchronization.

**8.** The method of claim 1 used in combination with a self-sustaining neural-motor energy harvesting loop, wherein the neural processing system's motor output generates electrical energy through piezoelectric and thermoelectric transduction, the autonomous fallback controller adjusts neural activity to maintain energy homeostasis, and the system can operate indefinitely without external power or cognitive control, limited only by the hard cap on autonomous steps.

**9.** The method of claim 1 used in combination with configurable recursive self-observation, wherein the neural processing system observes its own prior output through a reflection feedback pathway, the autonomous fallback controller adjusts the reflection coefficient based on energy level, reduced energy leads to reduced self-observation for conservation, and surplus energy leads to increased self-observation for exploration and learning.

**10.** The system of claim 2 wherein said autonomous input generator produces a circadian-like base signal with a configurable period defaulting to 500 steps and amplitude defaulting to 0.3, combined with stochastic attention bursts occurring with a configurable probability defaulting to 10 percent per step and amplitude range defaulting to 0.3 to 0.8.

**11.** The system of claim 2 further comprising a crash recovery mechanism wherein upon system restart after an unexpected termination, said system checks for the existence of a persistent state snapshot, if a valid snapshot exists said system restores its complete state from said snapshot, restored state includes neural network weights, membrane potentials, energy storage level, and autonomous operation count, and the system resumes operation from the restored state rather than initializing fresh, enabling continuity across system crashes with minimal state loss.

**12.** The system of claim 2 further comprising a clean shutdown mechanism wherein upon detection of process termination including stdin EOF or SIGTERM, the system executes a shutdown handler that serializes current state to persistent storage, enabling full state recovery on subsequent restart with zero data loss.

## ABSTRACT OF THE DISCLOSURE

A method and system for maintaining continuous operation of a neural processing system during disconnection from an external cognitive control layer. A heartbeat watchdog detects disconnection when a configurable timeout threshold is exceeded. Upon disconnection, an autonomous mode engages seamlessly, providing self-regulating input through a circadian-like base signal with stochastic attention bursts. An energy-aware modulation controller adjusts operational parameters based on current energy storage, implementing multi-threshold regulation that conserves resources when energy is low and enables exploration when surplus. Operational state is buffered for resynchronization and periodically serialized for crash recovery. Upon reconnection, a resynchronization protocol transmits summary statistics and optionally full operational history, restoring cognitive control without state loss. A hard cap limits maximum autonomous operation. The system targets environments where continuous operation is critical, including space exploration, prosthetics, and remote IoT sensors. (FIG. 7)

## REFERENCE TO SOURCE CODE

The complete implementation of this invention is available at the following repository: github.com/DarkWinD90/Consciousness_Env, validated state tagged as v3.0.0-phase10-multimodal at commit 9e2c333. Key implementation file is mcp/consciousness_mcp_server.py. Persistent state snapshots are stored in mcp/.snapshots/ directory.

## SUPPLEMENTARY APPLICATION CONTENT — PHASE 13 RECURRENT DEPTH

### PROVENANCE NOTE

This supplementary section was added on 2026-04-25 on branch claude/consciousness-recurrent-structure-KPrLB to capture the new claims introduced by Phase 13 (Recurrent Depth). It is part of the canonical GitHub specification; no external reconciliation is required. Claims 1 through 12 above are unchanged.

### PHASE 13 SUMMARY OF NEW ELEMENT

Phase 13 of the reference implementation adds an inner-loop wrapper around the spiking neural network's integrate-and-fire step (described in paragraphs 0017 through 0023 of the co-pending Patent B application). The autonomous runner described in paragraph 0030 of this application may step a recurrent-depth-enabled spiking neural network in place of a single-iteration spiking neural network. When so configured, each autonomous step invokes T inner iterations of the network's integrate-and-fire dynamics per autonomous step before the autonomous controller's state buffer records the step. The energy-aware modulation scheme described in paragraph 0022 is extended to optionally control the recurrent-depth parameter T as a function of energy state. The same multi-region energy curve that maps energy to a scalar modulation value can additionally select an inner-iteration depth: lower depth conserves neural compute when energy is constrained; higher depth permits deeper attractor convergence (and therefore richer reasoning) when energy is abundant. The buffer entries described in paragraph 0024 include the depth parameter T used at each autonomous step so that the resynchronization payload faithfully reports the compute regime during autonomous operation.

## ADDITIONAL CLAIMS (PROPOSED FOR INCORPORATION)

13. The method of claim 1 wherein said autonomous input generator provides input to a recurrent-depth-enabled spiking neural network that executes T inner iterations of integrate-and-fire dynamics per autonomous step, where T is a configurable integer greater than or equal to one, and wherein said operational state buffer records the value of T applied at each autonomous step alongside the input signal, modulation value, spike count, and energy level.

14. The method of claim 13 wherein said energy-aware regulation scheme additionally selects the value of T as a function of the current energy storage level, said selection reducing T toward one when energy is low and increasing T when energy is in surplus, such that inner-loop reasoning depth is itself an energy-aware operating parameter.

15. The system of claim 2 further comprising a recurrent-depth controller coupled between said autonomous input generator and said neural processing unit, wherein said recurrent-depth controller invokes the neural processing unit's integrate-and-fire step T times per autonomous step with input re-injection at each said inner iteration, and wherein the cumulative spike count produced across all T inner iterations is recorded as a single buffer entry per autonomous step.

## VALIDATION EVIDENCE FOR CLAIMS 13-15

The depth-aware extension is structurally compatible with the existing fallback runner: Phase 13 falsifiable claim F13.2 verifies that across 200 outer steps the energy harvester is charged exactly once per outer step regardless of T. This invariant is the precondition for embedding a recurrent-depth SNN inside the autonomous runner without breaking the operational state buffer's one-entry-per-step contract. Phase 13 falsifiable claim F13.1 (correlation ratio 1.0804 versus threshold 1.05) and claim F13.3 (variance ratio 0.0379 versus threshold 0.7) further establish that the recurrent-depth wrapper produces a more strongly input-coupled and more rapidly convergent hidden state than the single-iteration baseline.

**REFERENCE TO SOURCE CODE (PHASE 13)**

Repository: github.com/DarkWinD90/Consciousness_Env. Branch: claude/consciousness-recurrent-structure-KPrLB. Commit at time of supplement: aac99fc. Pending tag: v4.0.0-phase13-recurrent-depth (post-merge to main). Implementation file: core/recurrent_depth.py. Integration point for the autonomous runner: substitute RecurrentDepthSNN for BaseSNN in the runner's neural-processing dispatch (the wrapper preserves the one-step-result-per-outer-call contract).